Lecture 6: An Interior-point Algorithm

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Course of Introduction to Optimization

Fall, 2025

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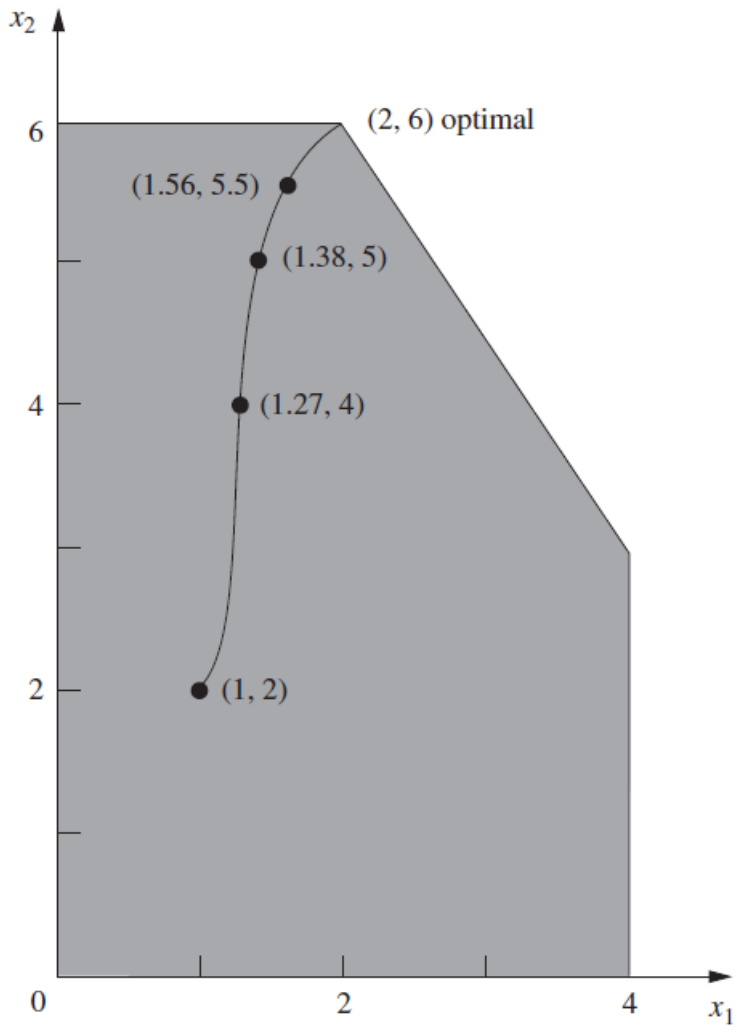
An Interior-point Algorithm

The ideas to be described can be summarized as follows:

Concept 1: Shoot through the interior of the feasible region toward an optimal solution.

Concept 2: Move in a direction that improves the objective function value at the fastest possible rate.

Concept 3: Transform the feasible region to place the current trial solution near its center, thereby enabling a large improvement when concept 2 is implemented.



An Interior-point Algorithm

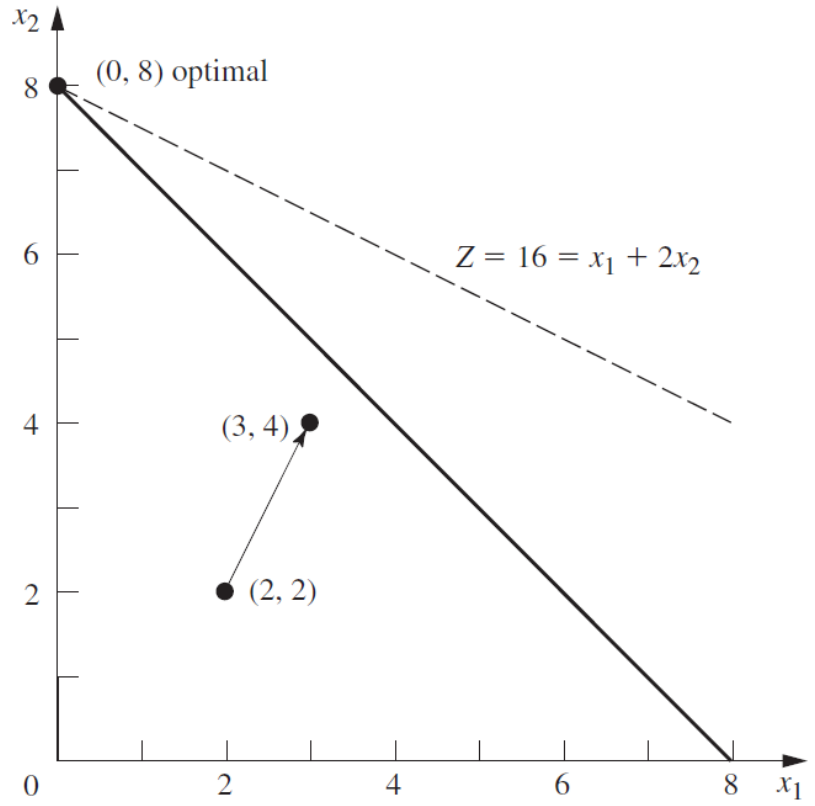
To illustrate these ideas throughout the section, we shall use the following example:

$$\text{Maximize } z = x_1 + 2x_2$$

Subject to

$$x_1 + x_2 \leq 8$$
$$x_1, x_2 \geq 0$$

This problem is depicted graphically in Fig., where the optimal solution is seen to be $(x_1, x_2) = (0,8)$ with $Z = 16$. (We will describe the significance of the arrow in the figure shortly.)



You will see that our interior-point algorithm requires a considerable amount of work to solve this tiny example.

The reason is that the algorithm is designed to solve huge problems efficiently, but is much less efficient than the simplex method (or the graphical method in this case) for small problems.

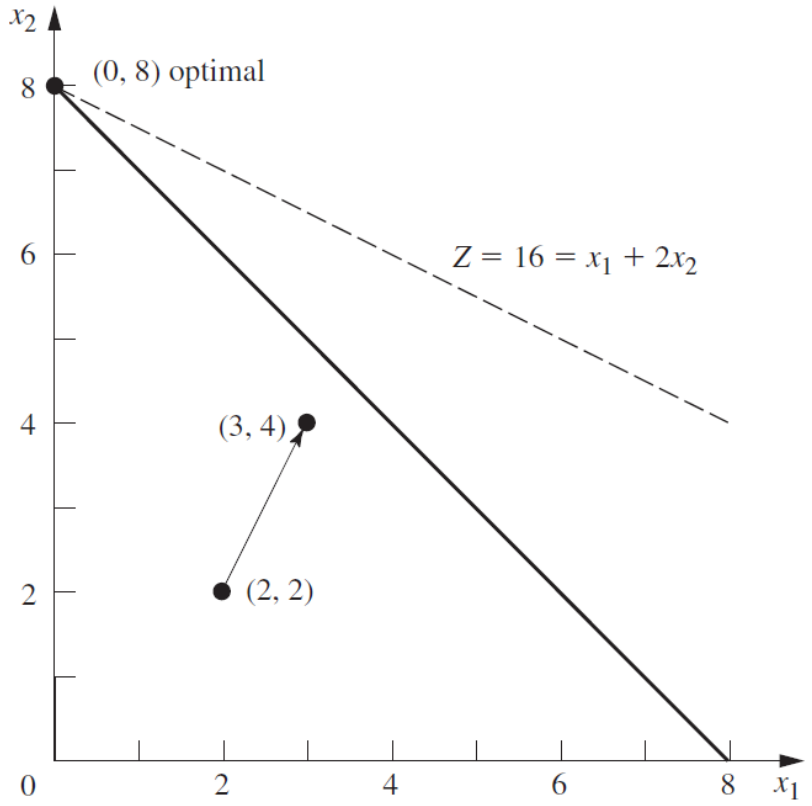
An Interior-point Algorithm

The algorithm begins with an initial trial solution that (like all subsequent trial solutions) lies in the interior of the feasible region, i.e., inside the boundary of the feasible region.

Thus, for the example, the solution must not lie on any of the three lines $x_1, = 0\ ,\ x_2 = 0, x_1 + x_2 = 8$

A trial solution that lies on the boundary cannot be used because this would lead to the undefined mathematical operation of division by zero at one point in the algorithm.

In this example, We have arbitrarily chosen be $(x_1, x_2) = (2,2)$ to be the initial trial solution.



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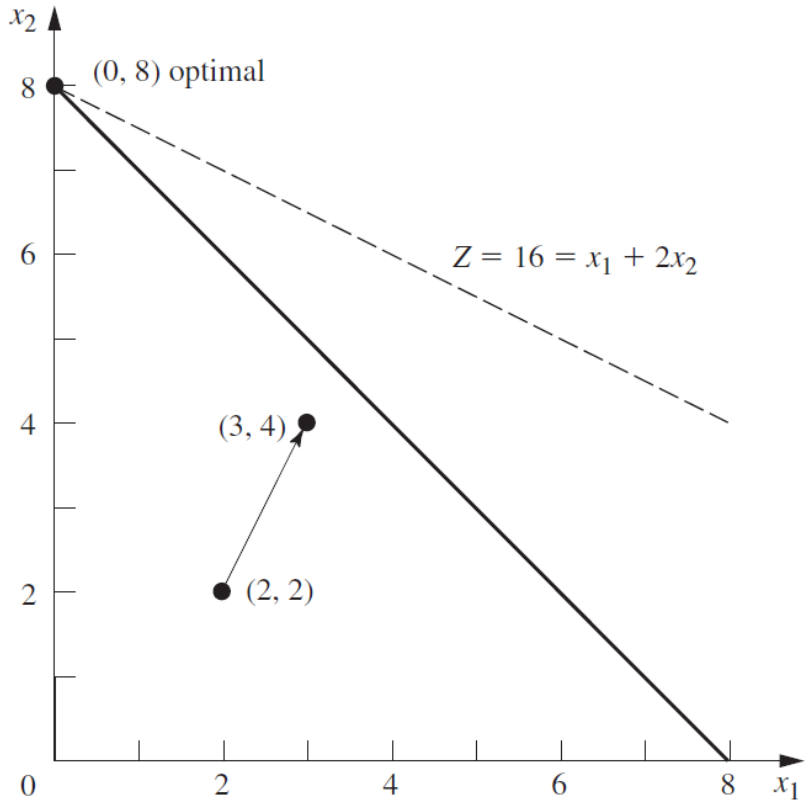
To begin implementing concepts 1 and 2, note in Fig. that the direction of movement from (2, 2) that increases Z at the fastest possible rate is perpendicular to (and toward) the objective function line $z = x_1 + 2x_2 = 16$.

We have shown this direction by the arrow from (2, 2) to (3, 4). Using vector addition, we have

$$(3,4) = (2,2) + (1,2),$$

where the vector (1, 2) is the **gradient** of the objective function.

The components of (1, 2) are just the coefficients in the objective function.



The gradient (1, 2) defines the ideal direction to which to move, where the question of the *distance to move*.

An Interior-point Algorithm

The algorithm actually operates on linear programming problems after they have been rewritten in augmented form. Letting x_3 be the slack variable for the functional constraint of the example, we see that this form is

$$\text{Maximize } z = x_1 + 2x_2$$

Subject to

$$x_1 + x_2 + x_3 = 8$$
$$x_1, x_2, x_3 \geq 0$$

The augmented form can be written in general as

$$\text{Maximize } z = \mathbf{c}^T\mathbf{x}$$

Subject to

$$A\mathbf{x} = \mathbf{b}$$
$$\mathbf{x} \geq 0$$

Where

$$c = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad A = [1 \quad 1 \quad 1], \qquad b = [8], \qquad \mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

An Interior-point Algorithm

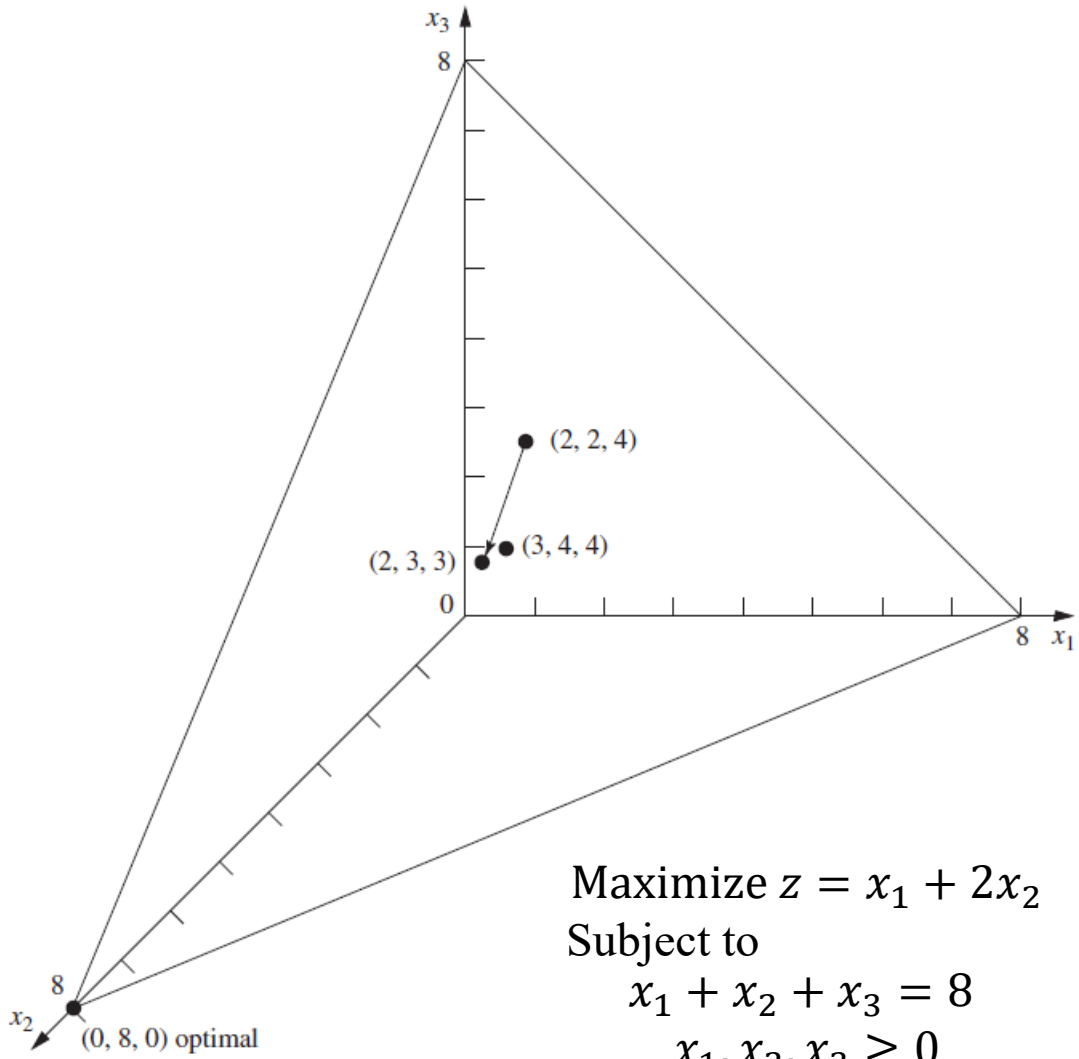
Note that $c^T = [1, 2, 0]$ now is the gradient of the objective function.

The augmented form of the example is depicted graphically in Fig.

The feasible region now consists of the triangle with vertices (8, 0, 0), (0, 8, 0), and (0, 0, 8).

Points in the interior of this feasible region are those where $x_1 > 0, x_2 > 0$, and $x_3 > 0$.

Each of these three $x_j > 0$ conditions has the effect of forcing (x_1, x_2) away from one of the three lines forming the boundary of the feasible region in fig in slide 3.



Maximize $z = x_1 + 2x_2$

Subject to

$x_1 + x_2 + x_3 = 8$

$x_1, x_2, x_3 \geq 0$

Using the Projected Gradient to Implement Concepts 1 and 2

In augmented form, the initial trial solution for the example is $(x_1, x_2, x_3) = (2, 2, 4)$.

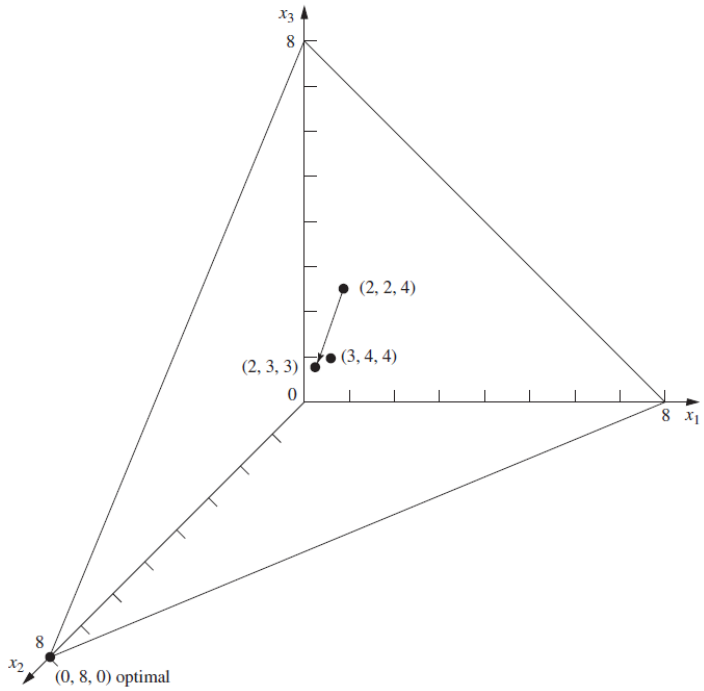
Adding the gradient (1, 2, 0) leads to

$$(3, 4, 4) = (2, 2, 4) + (1, 2, 0)$$

However, now there is a complication. The algorithm cannot move from (2, 2, 4) to (3, 4, 4), because (3, 4, 4) is infeasible!

When $x_1 = 3$ and $x_2 = 4$, then $x_3 = 8 - x_1 - x_2 = 1$ instead of 4.

The point (3, 4, 4) lies on the near side as you look down on the feasible triangle in Fig.



Maximize $z = x_1 + 2x_2$

Subject to

$$x_1 + x_2 + x_3 = 8$$

$$x_1, x_2, x_3 \geq 0$$

An Interior-point Algorithm

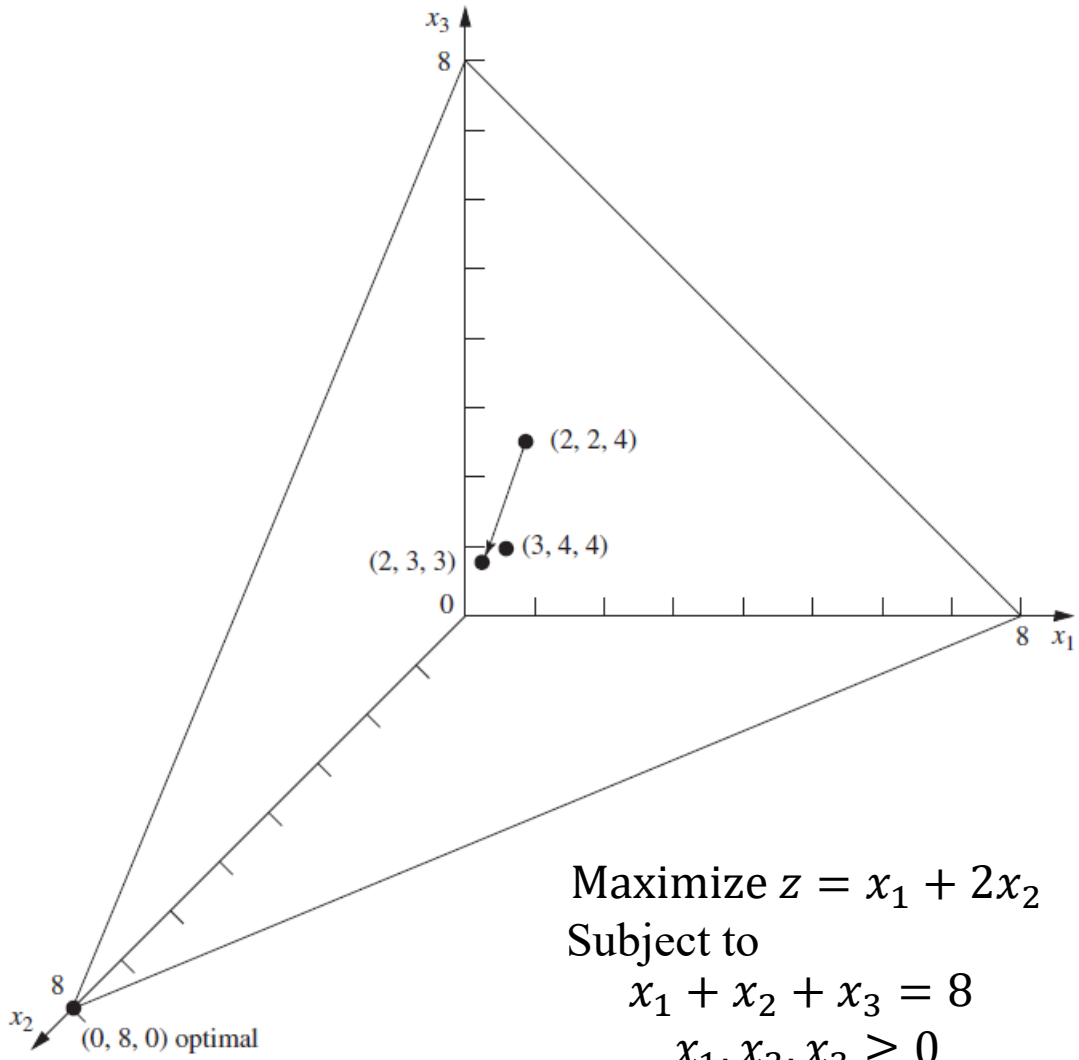
Therefore, to remain feasible, the algorithm (indirectly) projects the point (3, 4, 4) down onto the feasible triangle by dropping a line that is perpendicular to this triangle.

A vector from (0, 0, 0) to (1, 1, 1) is perpendicular to this triangle, so the perpendicular line through (3, 4, 4) is given by the equation

$$(x_1, x_2, x_3) = (3, 4, 4) - \theta(1, 1, 1),$$

where θ is a scalar.

Since the triangle satisfies the equation $x_1 + x_2 + x_3 = 8$, this perpendicular line intersects the triangle at (2, 3, 3).



Maximize $z = x_1 + 2x_2$

Subject to

$x_1 + x_2 + x_3 = 8$

$x_1, x_2, x_3 \geq 0$

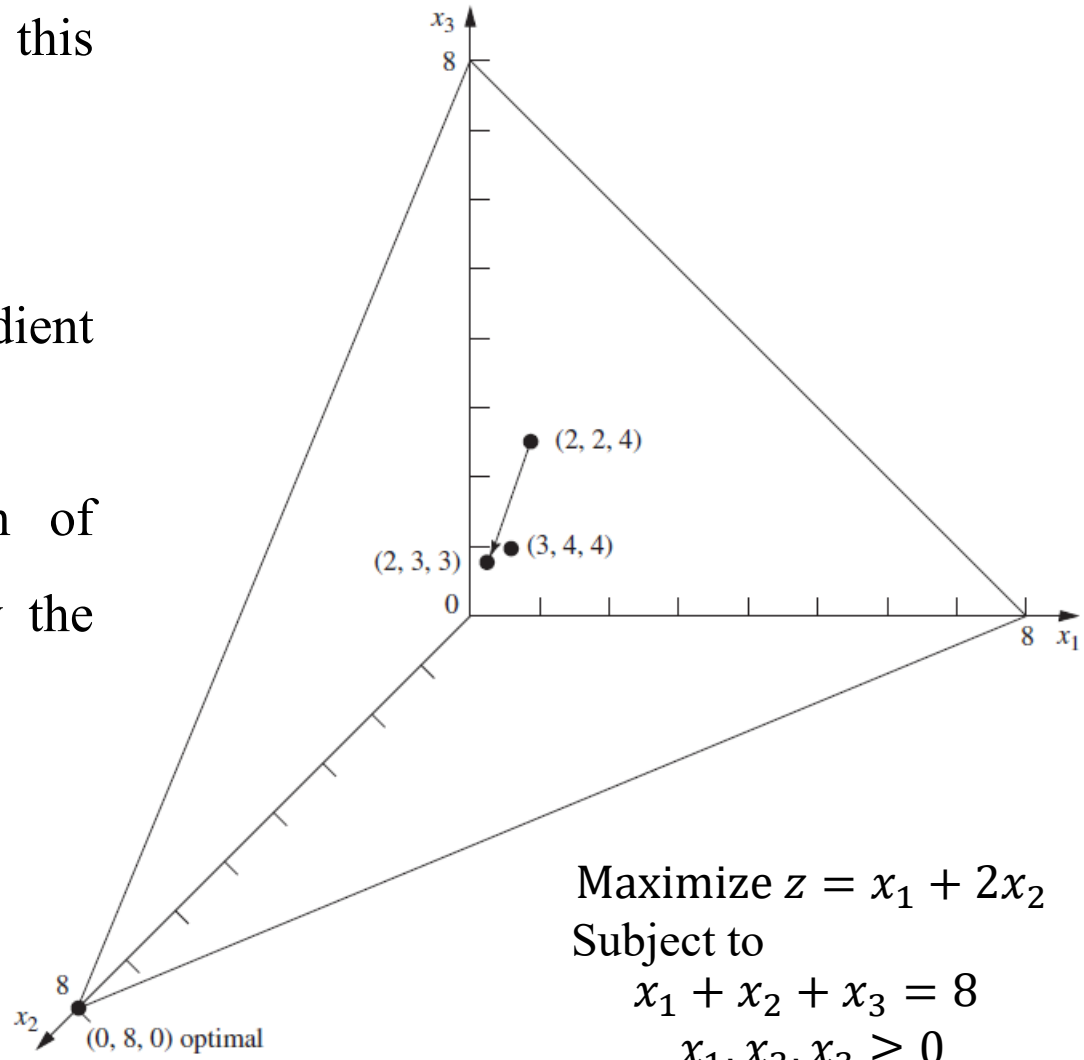
An Interior-point Algorithm

Since the triangle satisfies the equation $x_1 + x_2 + x_3 = 8$, this perpendicular line intersects the triangle at (2, 3, 3). Because

$$(2, 3, 3) = (2, 2, 4) + (0, 1, -1),$$

the projected gradient of the objective function (the gradient projected onto the feasible region) is (0, 1, -1).

It is this projected gradient that defines the direction of movement from (2, 2, 4) for the algorithm, as shown by the arrow in Fig.



Maximize $z = x_1 + 2x_2$

Subject to

$x_1 + x_2 + x_3 = 8$

$x_1, x_2, x_3 \geq 0$

A formula is available for computing the projected gradient directly. By defining the projection matrix P as

$$P = I - A^T(AA^T)^{-1}A$$

the projected gradient (in column form) is

$$c_p = Pc$$

Thus, for the example,

$$P = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \begin{bmatrix}1\\1\\1\end{bmatrix}\left(\begin{bmatrix}1 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}\right)^{-1}\begin{bmatrix}1 & 1 & 1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix}\begin{bmatrix}1 & 1 & 1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \frac{1}{3}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}2/3 & -1/3 & -1/3\\ -1/3 & 2/3 & -1/3\\ -1/3 & -1/3 & 2/3\end{bmatrix}$$

Moving from (2, 2, 4) in the direction of the projected gradient (0, 1, -1) involves increasing from zero in the formula

Moving from (2, 2, 4) in the direction of the projected gradient (0, 1, -1) involves increasing α from zero in the formula

$$x = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} + 4\,\alpha c_p = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} + 4\,\alpha \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

where the coefficient 4 is used simply to give an upper bound of 1 for α to maintain feasibility (all $x_j \geq 0$).

Note that increasing α to $\alpha = 1$ would cause x_3 to decrease to $x_3 = 4 + 4(1)(-1) = 0$, where $\alpha > 1$ yields $x_3 < 0$.

Thus, α measures the fraction used of the distance that could be moved before the feasible region is left.

An Interior-point Algorithm

How large should α be made for moving to the next trial solution?

Because the increase in Z is proportional to α, a value close to the upper bound of 1 is good for giving a relatively large step toward optimality on the current iteration.

However, the problem with a value too close to 1 is that the next trial solution then is jammed against a constraint boundary, thereby making it difficult to take large improving steps during subsequent iterations.

Therefore, it is very helpful for trial solutions to be near the center of the feasible region (or at least near the center of the portion of the feasible region in the vicinity of an optimal solution), and not too close to any constraint boundary.

With this in mind, Karmarkar has stated for his algorithm that a value as large as $\alpha = 0.25$ should be "safe." In practice, much larger values (for example, $\alpha = 0.9$) sometimes are used.

For the purposes of this example (and the problems at the end of the chapter), we have chosen $\alpha = 0.5$.

A Centering Scheme for Implementing Concept 3

The basic idea of the centering scheme is straightforward - simply change the scale (units) for each of the variables so that the trial solution becomes equidistant from the constraint boundaries in the new coordinate system.

For the example, there are three constraint boundaries in Fig. slide 3, each one corresponding to a zero value for one of the three variables of the problem in augmented form, namely, $x_1 = 0, x_2 = 0, \text{and } x_3 = 0$.

In Fig. slide 5, see how these three constraint boundaries intersect the $Ax = b$ $(x_1 + x_2 + x_3 = 8)$ plane to form the boundary of the feasible region.

The initial trial solution is $(x_1, x_2, x_3) = (2, 2, 4)$, so this solution is 2 units away from the , $x_1 = 0, x_2 = 0$ constraint boundaries and 4 units away from the $x_3 = 0$ constraint boundary, when the units of the respective variables are used.

However, whatever these units are in each case, they are quite arbitrary and can be changed as desired without changing the problem.

Therefore, let us rescale the variables as follows:

$$\tilde{x}_1 = \frac{x_1}{2}, \qquad \tilde{x}_2 = \frac{x_2}{2}, \qquad \tilde{x}_3 = \frac{x_3}{4}$$

in order to make the current trial solution of $(x_1, x_2, x_3) = (2, 2, 4)$ become

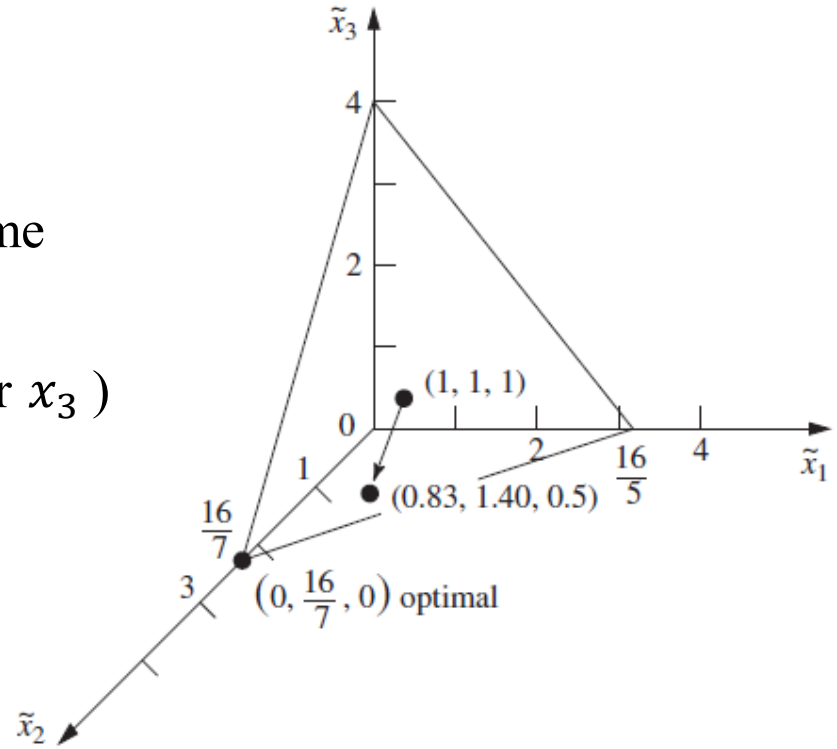
$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = (1,1,1)$$

In these new coordinates (substituting $2\tilde{x}_1$ for x_1, $2\tilde{x}_2$ for x_2 and $4\tilde{x}_3$ for x_3) the problem becomes

$$\text{Maximize } z = 2\tilde{x}_1 + 4\tilde{x}_2$$

Subject to

$$2\tilde{x}_1 + 2\tilde{x}_2 + 4\tilde{x}_3 = 8$$
$$\tilde{x}_i \geq 0, \qquad i = 1, 2, 3$$



Note that the trial solution (1, 1, 1) in Fig. is equidistant from the three constraint boundaries $\tilde{x}_1 = 0, \tilde{x}_2 = 0, \tilde{x}_3 = 0$.

For each subsequent iteration as well, the problem is rescaled again to achieve this same property, so that the current trial solution always is (1, 1, 1) in the current coordinates.

Summary and Illustration of the Algorithm

Iteration 1. Given the initial trial solution $(x_1, x_2, x_3) = (2, 2, 4)$, let D be the corresponding diagonal matrix such that $D\tilde{x} = x$, so that

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

The rescaled variables then are the components of

$$\tilde{x} = D^{-1}x = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1/2 \\ x_2/2 \\ x_3/4 \end{bmatrix}$$

In these new coordinates, A and c have become

$$\tilde{A} = AD = [1 \quad 1 \quad 1] \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} = [2 \quad 2 \quad 4]$$

$$\tilde{c} = Dc = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}$$

Therefore, the projection matrix is

$$P = I - \tilde{A}^T\left(\tilde{A}\tilde{A}^T\right)^{-1}\tilde{A}$$

$$P = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \begin{bmatrix}2\\ 2\\ 4\end{bmatrix}\left(\begin{bmatrix}2 & 2 & 4\end{bmatrix}\begin{bmatrix}2\\ 2\\ 4\end{bmatrix}\right)^{-1}\begin{bmatrix}2 & 2 & 4\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \frac{1}{24}\begin{bmatrix}4 & 4 & 8\\ 4 & 4 & 8\\ 8 & 8 & 16\end{bmatrix} = \begin{bmatrix}5/6 & -1/6 & -1/3\\ -1/6 & 5/6 & -1/3\\ -1/3 & -1/3 & 1/3\end{bmatrix}$$

so that the projected gradient is

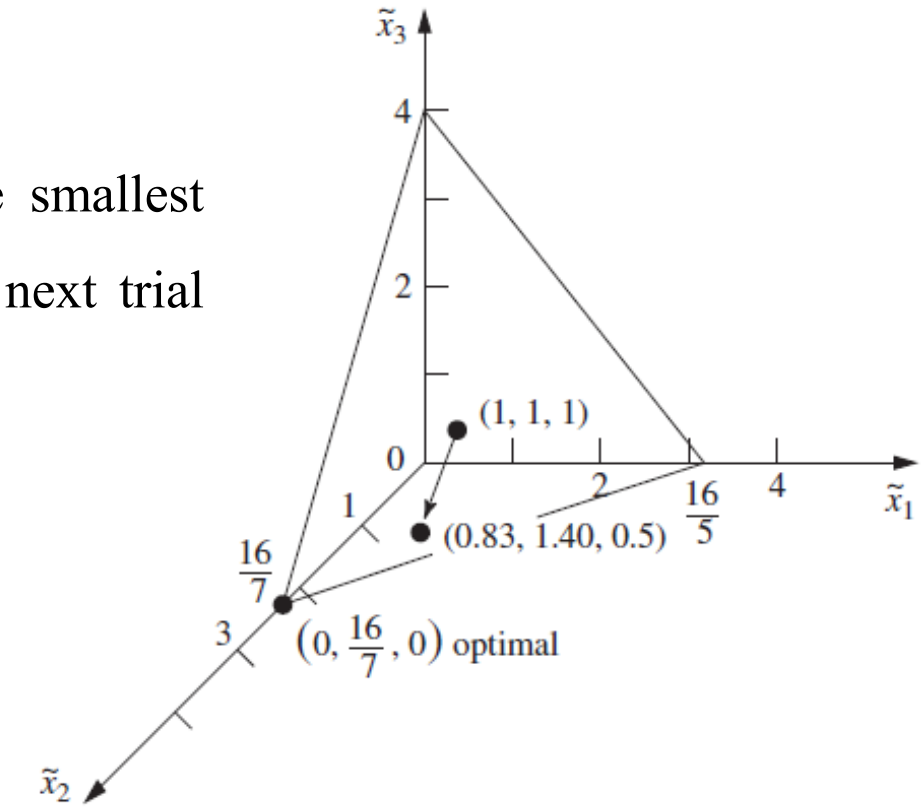
$$c_p = P\tilde{c} = \begin{bmatrix}5/6 & -1/6 & -1/3\\ -1/6 & 5/6 & -1/3\\ -1/3 & -1/3 & 1/3\end{bmatrix}\begin{bmatrix}2\\ 4\\ 0\end{bmatrix} = \begin{bmatrix}1\\ 3\\ -2\end{bmatrix}$$

Define v as the absolute value of the negative component of c_p having the largest absolute value, so that $v = |-2| = 2$ in this case.

Consequently, in the current coordinates, the algorithm now moves from the current trial solution $(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = (1, 1, 1)$ to the next trial solution

$$\tilde{x} = \begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\alpha}{v} c_p = \begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{0.5}{2}\begin{bmatrix}1\\3\\-2\end{bmatrix} = \begin{bmatrix}5/4\\7/4\\1/2\end{bmatrix}$$

as shown in Fig. The definition of v has been chosen to make the smallest component of $\tilde{x}$ equal to zero when $\alpha = 1$ in this equation for the next trial solution.



In the original coordinates, this solution is

$$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = D\tilde{x} = \begin{bmatrix}2 & 0 & 0\\0 & 2 & 0\\0 & 0 & 4\end{bmatrix}\begin{bmatrix}5/4\\7/4\\1/2\end{bmatrix} = \begin{bmatrix}5/2\\7/2\\2\end{bmatrix}$$

This completes the iteration, and this new solution will be used to start the next iteration.

These steps can be summarized as follows for any iteration.

Summary of the Interior-Point Algorithm

1. Given the current trial solution $(x_1, x_2, \ldots, x_n)$, set

$$D = \begin{bmatrix} x_1 & 0 & 0 & \cdots & 0 \\ 0 & x_2 & 0 & \cdots & 0 \\ 0 & 0 & x_3 & \cdots & 0 \\ \cdots & \cdots & \cdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & x_n \end{bmatrix}$$

2. Calculate $\tilde{A} = AD$ and $\tilde{c} = Dc$

3. Calculate $P = I - \tilde{A}^T\left(\tilde{A}\tilde{A}^T\right)^{-1}\tilde{A}$ and $c_p = P\tilde{c}$

4. Identify the negative component of c_p having the largest absolute value, and set v equal to this absolute value. Then calculate

$$\tilde{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \frac{\alpha}{v} c_p$$

where α is a selected constant between 0 and 1 (for example, $\alpha = 0.5$).

5. Calculate $x = D\tilde{x}$ as the trial solution for the next iteration (step 1). (If this trial solution is virtually unchanged from the preceding one, then the algorithm has virtually converged to an optimal solution, so stop.)

Iteration 2

Step 1:

Given the current trial solution (x_1, x_2, x_3) , set

$$D = \begin{bmatrix} 5/2 & 0 & 0 \\ 0 & 7/2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

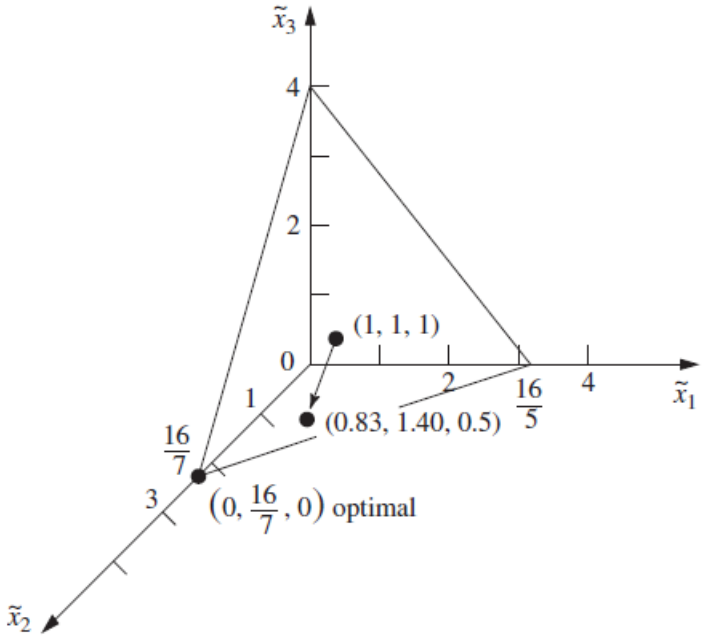
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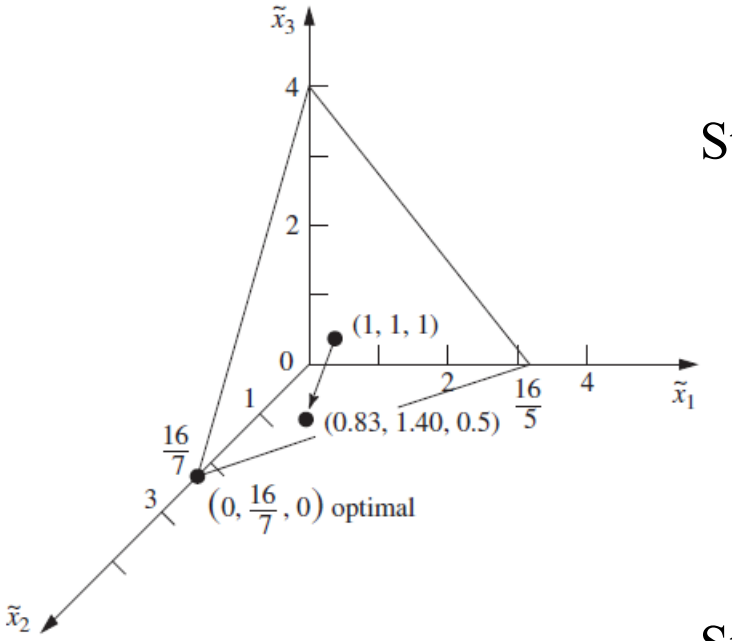
$$\begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix} = D^{-1}x = \begin{bmatrix} 2/5 & 0 & 0 \\ 0 & 2/7 & 0 \\ 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1/5 \\ 2x_2/7 \\ x_3/2 \end{bmatrix}$$

so that the BF solutions in these new coordinates are

$$\tilde{x} = D^{-1}\begin{bmatrix} 8 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{16}{5} \\ 0 \\ 0 \end{bmatrix}, \qquad \tilde{x} = D^{-1}\begin{bmatrix} 0 \\ 8 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{16}{7} \\ 0 \end{bmatrix}, \qquad \tilde{x} = D^{-1}\begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

as depicted in Fig.





Step 2:

$$\tilde{A} = AD = \begin{bmatrix} \frac{5}{2} & \frac{7}{2} & 2 \end{bmatrix}, \tilde{c} = DC = \begin{bmatrix} \frac{5}{2} \\ 7 \\ 0 \end{bmatrix}$$

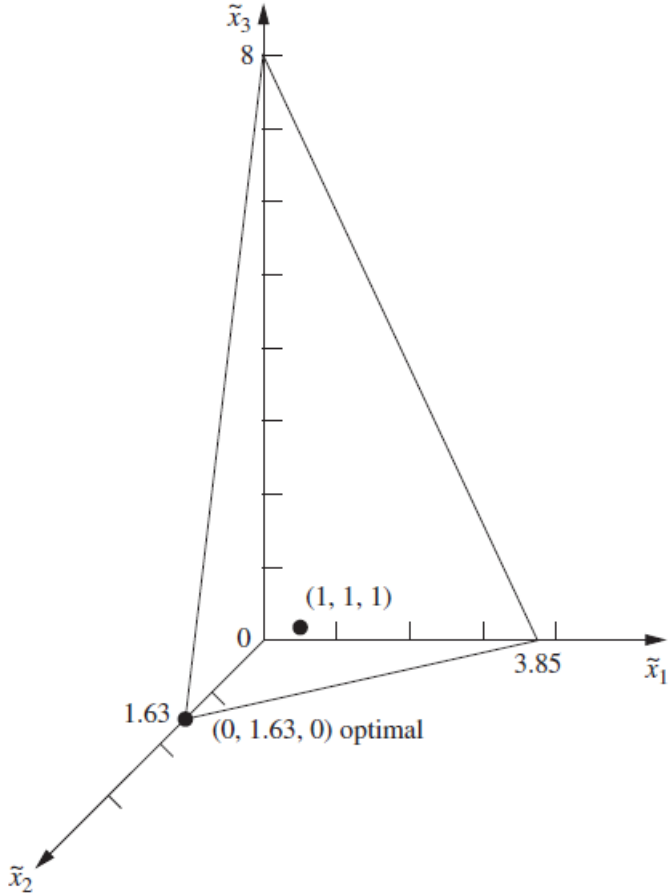
Step 3:

$$P = \begin{bmatrix} 13/8 & -7/18 & -2/9 \\ -7/18 & 41/90 & -14/45 \\ -2/9 & -14/45 & 37/45 \end{bmatrix} \text{ and } c_p = \begin{bmatrix} -11/12 \\ 133/60 \\ -41/15 \end{bmatrix}$$

Step 4:

$$\left| -\frac{41}{15} \right| > \left| -\frac{11}{12} \right|, \text{so } v = \frac{41}{15}$$

$$\tilde{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \frac{0.5}{\frac{41}{15}} \begin{bmatrix} -11/12 \\ 133/60 \\ -41/15 \end{bmatrix} \approx \begin{bmatrix} 0.83 \\ 1.4 \\ 0.5 \end{bmatrix}$$



Step 5:

$$x = D\tilde{x} = \begin{bmatrix} 1365/656 \\ 3227/656 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 2.08 \\ 4.92 \\ 1 \end{bmatrix}$$

is the trial solution for iteration 3.

Since there is little to be learned by repeating these calculations for additional iterations, we shall stop here.

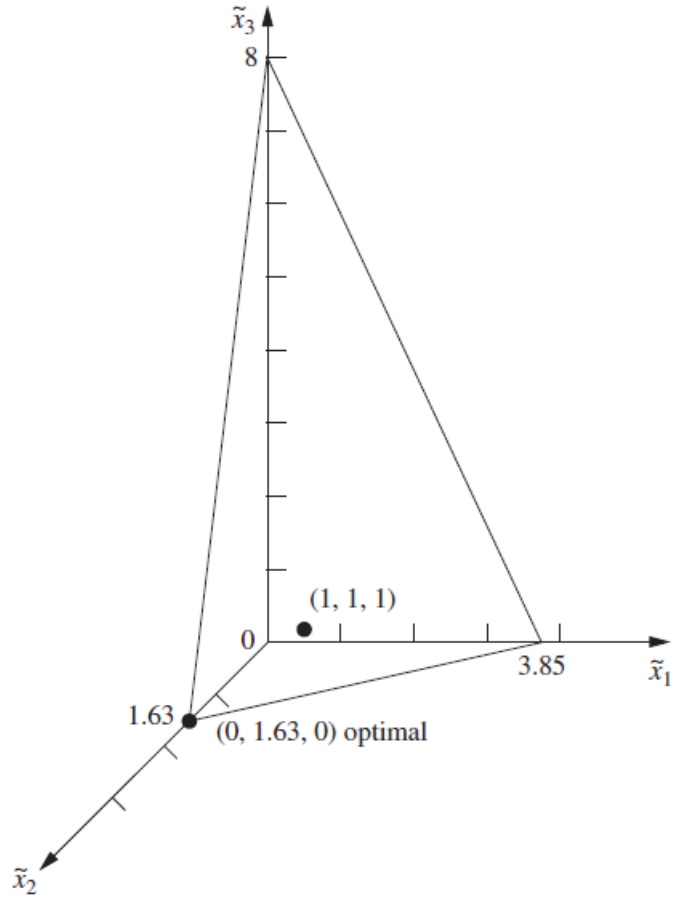


Figure shows the progress of the algorithm in the original x_1 x_2 coordinate system before the problem is augmented.

The three points (x_1, x_2) $(2, 2), (2.5, 3.5),$ and $(2.08, 4.92)$ are the trial solutions for initiating iterations 1, 2, and 3, respectively.

We then have drawn a smooth curve through and beyond these points to show the trajectory of the algorithm in subsequent iterations as it approaches $(x_1, x_2) = (0, 8)$.

